

# To Memorize or to Retrieve: Scaling the Interaction Between Pretraining and Retrieval

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## Abstract

Retrieval-augmented generation (RAG) improves language model (LM) performance by providing relevant context at test time for knowledge-intensive situations. In this work, we systematically study the trade-off between pretraining and retrieval by training OLMo-2-based LMs ranging from 30M to 3B parameters on up to 100B DCLM tokens, while varying pretraining data scale, retrieval store size, and retrieval store source (pretraining vs. new data) across reasoning, scientific QA, and open-domain QA benchmarks. We find that retrieval gains depend on model capacity and pretraining exposure and are strongly front-loaded, with a median 91% of the largest observed improvement realized by one retrieval token per model parameter. However, the interaction is objective-dependent: smaller models gain more in gold-answer perplexity, whereas larger, more-pretrained models gain more in accuracy. Retrieval from previously seen data also preserves most of the held-out retrieval gain. Retrieval is therefore a task-, regime-, and metric-dependent complement to parametric learning whose value also depends on datastore size and information novelty. Overall, this motivates the explicit partitioning of data between internalization and external access for LM design.

## 1 Introduction

Scaling laws (Hestness et al., 2017; Kaplan et al., 2020) have established how language model (LM) performance improves with parameters and training tokens, but they treat the training corpus as monolithic. In standard pretraining, all available data is consumed parametrically, implicitly assuming that knowledge should be compressed into

model weights. Retrieval-augmented generation (RAG) introduces a new degree of freedom: a portion of the corpus can instead be held out as an external datastore and accessed at inference time. These two uses of data are fundamentally different, with distinct computational costs, inductive biases, and failure modes. For example, parametric learning may lead to an inaccurate internal world model and hallucination tendencies (Liu et al., 2026a,b), while RAG may lead to errors from retrieving irrelevant or misleading documents (Lewis et al., 2021). There are also ties to cognition: people typically internalize abstract reasoning skills while relying on external memory (e.g., books, search engines, notes) for factual recall (Wegner, 1987; Risko and Gilbert, 2016; Sparrow et al., 2011; Norman, 1988; Clark and Chalmers, 1998). While we do not directly study these mechanisms, they motivate viewing parametric and non-parametric knowledge as complementary resources, raising the question of how the value of external access changes as the generator acquires more parametric knowledge.

We study the question of how to allocate data between parametric knowledge and external knowledge through controlled experiments. Concretely, we draw both pretraining and retrieval data from the same DCLM corpus. For each model size, we vary the number of tokens used for pretraining  $D$  and placed in the retrieval datastore  $R$ . In our main condition, these sets are disjoint: increasing  $R$  gives the model inference-time access to information it did not encounter during pretraining. We also construct datastores from the model’s own pretraining data, testing whether retrieval remains useful when it provides easier access to previously seen information rather than new information. We fit a scaling law over model size, pretraining tokens, and datastore size, and use it to compare pretraining and retrieval within controlled token-budget slices.

Pretraining internalizes information in model parameters, while retrieval makes corpus information

Code for our paper is available at <https://github.com/DegenAI-Labs/RAG-Scaling-Laws>. Correspondence to karanps@stanford.edu, vgtomahawk@gmail.com, and syfeng@stanford.edu.

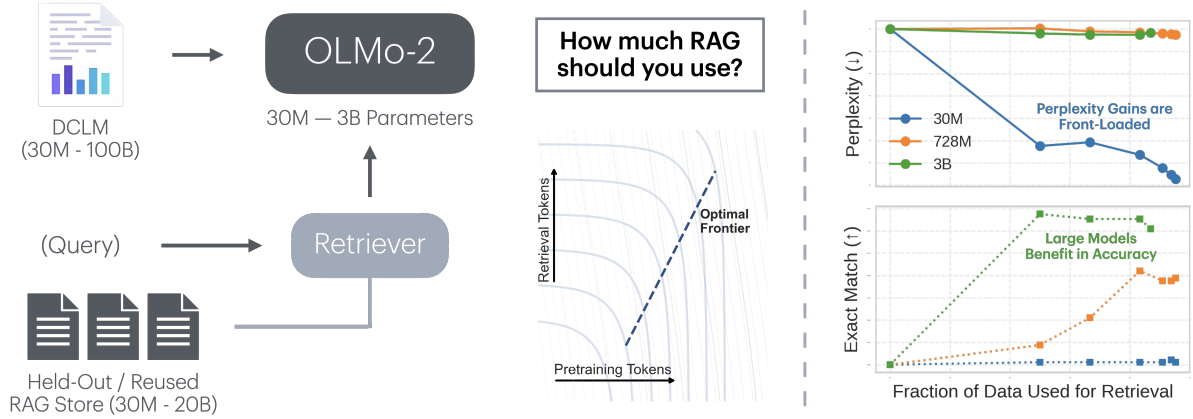


Figure 1: **Overview of the controlled pretraining–retrieval scaling study.** *Left:* We train OLMo-2 models from 30M to 3B parameters while varying DCLM pretraining exposure, datastore size, and whether indexed data is held out from or reused during pretraining. *Center:* We fit a scaling law over model size, pretraining tokens, and retrieval tokens; fixed-token-budget slices support performance-side comparisons among allocations. *Right:* The interaction with model scale depends on the evaluation target. Gold-answer perplexity gains are front-loaded and larger for smaller models, whereas decision-accuracy gains favor larger models on the multiple-choice benchmarks studied.

available as context at inference time. Understanding the tradeoff between these two mechanisms is essential for designing efficient, scalable language model systems. We study this empirically across model scales ranging from 30M to 3B parameters, systematically varying the amount of pretraining data and the size of the retrieval datastore constructed from the same corpus (Figure 1). We evaluate across a diverse set of benchmarks spanning multiple domains and knowledge types. In summary, we make the following contributions:

- We introduce an interaction scaling law and show that retrieval gains depend on model size, pretraining exposure, and evaluation target: smaller, less-pretrained models gain more in gold-answer likelihood, while larger, more-pretrained models gain more in choice-normalized probability and accuracy.
- This distinction separates likelihood scaling, which measures support for the correct answer, from decision-level scaling, which is relevant when optimizing downstream accuracy.
- We show that retrieval gains are strongly front-loaded: a median 91% of the largest observed gain is realized with one datastore token per model parameter, followed by sharply diminishing improvements at larger datastore sizes.
- By comparing held-out and reused datastores, we separate the benefit of accessing novel information from that of re-accessing information seen during pretraining. Reuse preserves 72–92% of the held-out retrieval gain.

Overall, we show that pretraining and retrieval cannot be analyzed independently: retrieval’s value depends on model capacity, pretraining exposure, task, and whether the indexed information was previously seen. Our scaling framework provides a controlled account of when external access complements or substitutes for additional pretraining, motivating a shift in how pretraining corpora should be conceptualized and helping to inform LM design.

## 2 Related Works

### 2.1 Scaling Laws for Pretraining

Many works study how LM performance scales with model size, dataset size, and compute. Kaplan et al. (2020) established predictable power-law relationships between these factors. Chinchilla later showed that compute-optimal training requires jointly scaling model and data size (Hoffmann et al., 2022). More recent work extended this: Gadre et al. (2024) show that scaling laws remain predictive in overtrained regimes and relate pretraining loss to downstream task performance, while other work incorporates data mixture (Ye et al., 2025; Shukor et al., 2025) and domain-specific continual pretraining (Que et al., 2024).

These works suggest that pretraining efficiency depends not only on parameters and tokens, but also on data and its proportions. A related line of work studies scaling when inference cost matters. Sardana et al. (2024) show that smaller models trained longer can be preferable when inference demand is high. Bian et al. (2025) incorporate

architecture-aware latency into scaling analysis, showing that parameter count alone is an incomplete proxy for efficiency. These results motivate treating pretraining, model size, data quality, and cost as coupled optimization problems.

## 2.2 Retrieval-Augmented Language Models

Retrieval-augmented LMs address a key limitation of purely parametric LMs: knowledge is stored implicitly in weights, making updates expensive and provenance difficult to trace. REALM was among the first to integrate retrieval directly into pretraining by jointly learning a dense retriever with a masked-LM objective (Guu et al., 2020). RAG popularized retrieval-augmented generation for knowledge-intensive tasks by conditioning a generator on retrieved Wikipedia passages (Lewis et al., 2021), while kNN-LM showed that nearest-neighbor lookup can improve perplexity and domain adaptation (Khandelwal et al., 2020).

Subsequent work scaled this to larger corpora and general-purpose LMs. RETRO demonstrated that large external datastores can match larger parametric models (Borgeaud et al., 2022), and Atlas showed strong few-shot performance with retrieval-augmented models that can be updated independently of the generator (Izacard et al., 2022). Recent surveys emphasize trade-offs among retriever quality, memory freshness, grounding, and system complexity (Hu and Lu, 2025). While much of the RAG literature focuses on improving downstream factuality at test time, systems such as REALM, RETRO, and Atlas suggest that retrieval can alter the pretraining trade-off itself by offloading knowledge from parameters into external memory.

## 2.3 Small Language Models: Pretraining, Evaluation & Data Efficiency

Recent work on small language models (SLMs) emphasizes that performance under tight parameter budgets depends heavily on architecture, data quality, and training duration. TinyLlama and SmolLM2 show that prolonged pretraining and careful data curation can make  $\sim 1\text{B}$ -parameter models competitive with larger open baselines (Zhang et al., 2024; Allal et al., 2025). At smaller data budgets, BabyLM studies learning from only 10M–100M tokens (Warstadt et al., 2023; Hu et al., 2024a), motivating analyses of inductive biases and linguistic asymmetries (Kallini et al., 2024; Hu et al., 2025; Feng et al., 2026; Zeng et al., 2026).

Evaluation is especially consequential for SLMs.

HELM advocates multi-metric, scenario-based evaluation (Liang et al., 2023), while DataComp-LM combines standardized pretraining recipes with broad downstream evaluation (Li et al., 2025). Data-centric studies further show that specialized corpora can strengthen compact models (Gunasekar et al., 2023; Penedo et al., 2024; Allal et al., 2025), while concept, knowledge, and visual augmentation can improve commonsense generation (Lin et al., 2020; Feng et al., 2021, 2022, 2023).

Under fixed training budgets, deduplication (Lee et al., 2022), educational filtering (Penedo et al., 2024), domain reweighting (Xie et al., 2023), and mixture optimization (Ye et al., 2025) can improve data efficiency. Related work studies data mixtures across pretraining stages (Feng et al., 2024a; Liu et al., 2026c), mixtures of code and target-domain data (Ma et al., 2023; Baek et al., 2026), and data ordering or progressive model scaling curricula (Feng et al., 2024b; Singh et al., 2026). Together, these works show that data efficiency depends on what, when, and how information is presented.

## 2.4 Retrieval-Scaling

Prior work establishes that retrieval gains persist across datastore, model, and inference-compute scales. Shao et al. (2024) jointly vary generator size, pretraining exposure, and datastore size, showing that retrieval can shift compute-optimal frontiers. Fang et al. (2025) retrieve from models’ own pretraining corpora and show that previously encountered data retains test-time value, while Yue et al. (2025); Zhou et al. (2026) characterize how performance changes with retrieval and generation compute at inference time. Together, these studies demonstrate that both new and previously seen information can improve pretrained models.

**Summary and Motivation.** Prior work shows that language-model performance scales with pretraining resources, datastore size, and inference-time computation, and that even previously encountered data can retain value when retrieved at test time. What remains unclear is how retrieval’s marginal value changes as a model acquires more parametric knowledge, and how much of that value depends on whether the retrieved information was seen during pretraining. We address this gap through a controlled factorial study: we train matched OLMo-2 models from scratch across model and pretraining scales, vary datastore size, and compare held-out with reused retrieval data un-

der the same corpus and retrieval pipeline. This design lets us measure how external access to knowledge interacts with model capacity and pretraining saturation and quantify its likelihood-equivalent substitution for additional pretraining.

### 3 Methods

#### 3.1 Experimental Setup

For our experiments, we use the OLMo-2 series (OLMo et al., 2024) of LMs due to its strong empirical performance, alignment with open research practices, and modern architectural design. We define our own OLMo-2 model sizes and pretrain them across various scales: 30M, 136M, 233M, 728M, 1B, and 3B parameters (hyperparameter details in Appendix A.1). We use 100B tokens of DCLM data as our pretraining corpus (Li et al., 2025). We train all models using AdamW with a  $3 \times 10^{-4}$  peak learning rate ( $lr$ ),  $\beta_1 = 0.9$ ,  $\beta_2 = 0.95$ , and 0.1 weight decay. We adopt a warmup-stable-decay (WSD) schedule (Hu et al., 2024b) with 10% linear warmup (capped at 2k steps), a stable phase, and 10% linear decay to a minimum  $lr$  of  $6e^{-5}$ . Models are evaluated every 2k steps.

#### 3.2 Index Construction

We construct retrieval indices across multiple scales (1B–20B tokens) via FAISS (Douze et al., 2025) from both a held-out slice of DCLM and the non-held out slice that models were trained on by first computing per-chunk token counts over the embedding store, and then selecting chunks via a seeded random permutation. For each target budget (e.g., 30M, 60M, etc.), we take the shortest prefix of that permutation whose cumulative token count meets or slightly exceeds the target, then materialize the corresponding chunk texts/metadata and build a FAISS index over the selected embeddings. Because all budgets are prefixes of the same permutation (for fixed source data, filtering config, and seed), smaller-budget indices are strict subsets of larger-budget indices (e.g.,  $30M \subset 60M$ ), enabling controlled scaling comparisons where corpus size is the primary varying factor. For our index construction, we chose Qwen3-Embedding-8B from amongst 4 candidate choices on the basis of recall, and IVPFQ as the indexing algorithm. More index construction details are in Appendix A.2.

#### 3.3 Evaluation Protocol

We evaluate all models using a retrieval-augmented variant of EleutherAI’s *lm-evaluation-harness* (Gao et al., 2024), the *RAG-Evaluation-Harness* framework (Shao et al., 2024), across multiple benchmarks spanning reasoning, scientific QA, and open-domain QA: AI2-ARC (Easy and Challenge) (Clark et al., 2018), HellaSwag (Zellers et al., 2019), OpenBookQA (Mihaylov et al., 2018), SciQ (scientific QA) (Welbl et al., 2017), Natural Questions (Kwiatkowski et al., 2019), StrategyQA (Geva et al., 2021), SimpleQA (Wei et al., 2024), and CommonsenseQA (Talmor et al., 2019).

**RAG evaluation setup.** We retrieve the top- $k$  passages ( $k = 5$ , chosen via a small pilot sweep as a trade-off between retrieval quality and context budget) from a fixed FAISS index. Retrieved passages are concatenated as context, followed by the original question and answer choices (if applicable). The retriever is frozen and shared across all evaluations to isolate the effect of retrieval scale.

**Metrics.** We evaluate retrieval at three levels. For our scaling analysis, we use the character-normalized negative log-likelihood of the gold answer continuation, conditioned on the task prompt and, for RAG, the retrieved context. We report its exponentiated value as conditional gold-answer perplexity (PPL). This continuous measure captures changes in the probability assigned to the correct answer and is better suited to fitting scaling relationships (Tay et al., 2021; Krajewski et al., 2025). Second, for multiple-choice tasks, we normalize answer-choice likelihoods into a distribution  $q$  and compute multiclass Brier loss,  $\sum_j (q_j - \mathbf{1}[j = y])^2$ . Third, decision error records whether the highest-scoring choice is incorrect (scaling fits based directly on accuracy / error-rate are included in Appendix A.4). These targets form a ladder from absolute gold-answer support, through support relative to competing answers, to the discrete decision. We use gold-answer perplexity for the primary scaling and token-allocation analyses because it varies smoothly across all benchmarks, and use Brier loss and error rate to determine whether its interactions transfer to downstream decisions.

### 4 Experimental Results

#### 4.1 Parametric Scaling Baselines

We begin by establishing parametric scaling baselines in the absence of retrieval (retrieval index



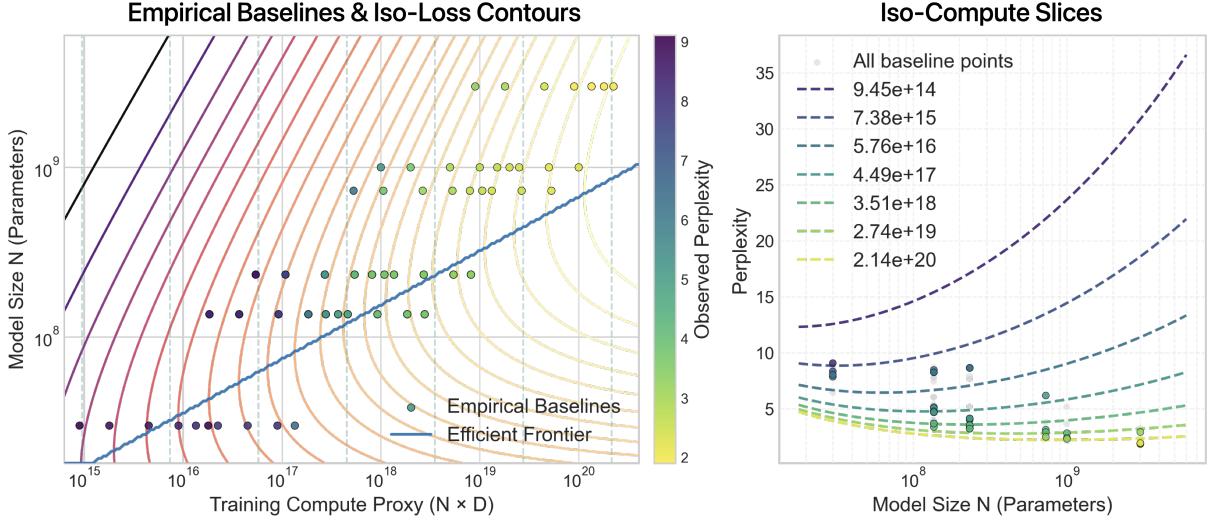


Figure 2: Parametric scaling baselines without RAG ( $R = 0$ ). *Left*: Empirical measurements across model sizes and data budgets, overlaid with iso-loss contours from the power-law model. Each point corresponds to a trained model configuration, colored by observed perplexity. The blue line denotes the compute-efficient frontier and the vertical dashed lines, discrete training budgets. *Right*: Iso-compute slices of the scaling surface, showing predicted loss as a function of model size ( $N$ ). Empirical observations are overlaid for reference.

|                    | Benchmark          | CV ARE (%) | LOMO ARE (%) | $\alpha$ | $\beta$ | $L_0$  |
|--------------------|--------------------|------------|--------------|----------|---------|--------|
| Reasoning          | CommonsenseQA      | 6.87       | 10.76        | 0.1558   | 0.0869  | 2.1619 |
|                    | HellaSwag          | 2.18       | 4.94         | 0.3923   | 0.4713  | 1.3714 |
|                    | StrategyQA         | 12.28      | 14.93        | 2.0000   | 0.4647  | 7.6735 |
| Scientific/Open QA | SciQ               | 13.07      | 24.03        | 0.5267   | 0.2606  | 0.9522 |
|                    | OpenBookQA         | 4.78       | 9.41         | 0.3688   | 0.2120  | 1.6579 |
|                    | AI2C-ARC Easy      | 7.10       | 13.67        | 0.3566   | 0.2195  | 1.0299 |
|                    | AI2C-ARC Challenge | 3.93       | 6.09         | 0.2811   | 0.2817  | 1.1133 |
|                    | Natural Questions  | 7.19       | 17.39        | 0.3595   | 0.3523  | 1.2593 |
|                    | SimpleQA           | 11.46      | 21.09        | 0.2285   | 0.9580  | 1.7196 |

Table 1: Power-law fit quality and scaling exponents for parametric baselines ( $R = 0$ ). We report cross-validation average relative error (CV ARE) [interpolation error under random splits] and leave-one-model-size-out ARE (LOMO ARE) [extrapolation error to unseen model size]. Lower ARE indicate better fit quality. Exponents  $\alpha$  and  $\beta$ , when combined with  $L_0$ , summarize how loss scales with model size and pretraining data in the baseline regime. ARE should be interpreted relative to the inherent noise and discreteness of benchmarks, with smoother, likelihood-based tasks yielding low errors, and reasoning-heavy tasks showing higher variance and ARE.

size  $R = 0$ ), varying model size ( $N$ ) and pretraining data ( $D$ ). This serves as a sanity check that our experimental setup reproduces the standard scaling-law behavior observed in prior work such as (Hoffmann et al., 2022). Following (Hoffmann et al., 2022), we model loss for parametric models as a function of model size and data using a power-law:

$$L(N, D) = A \left( \frac{N}{10^9} \right)^{-\alpha} + B \left( \frac{D}{10^9} \right)^{-\beta} + L_0 \quad (1)$$

where  $(A, \alpha)$  capture scaling with model size,  $(B, \beta)$  capture scaling with data, and  $L_0$  is an irreducible loss floor. Here,  $(A, \alpha)$  govern the model-size contribution,  $(B, \beta)$  govern the data contribu-

tion, and  $L_0$  is the asymptotic loss floor. Intuitively, larger  $\alpha$  implies stronger sensitivity to model scaling, while larger  $\beta$  implies stronger sensitivity to data scaling. Across benchmarks, as seen in Figure 2 and Table 1, we observe smooth and predictable improvements as either  $N$  or  $D$  increases, with diminishing returns in both directions as expected from power-law scaling. The fitted model achieves low average relative error:

$$\text{ARE} = \frac{1}{n} \sum_{i=1}^n |(\mathcal{L}_i^{\text{pred}} - \mathcal{L}_i^{\text{obs}}) / \mathcal{L}_i^{\text{obs}}| \times 100\%$$

and scaling exponents broadly align with previously reported values (Hoffmann et al., 2022).

Overall, this validates that our setup reliably reproduces canonical scaling-law behavior.

## 4.2 Scaling Laws for Retrieval

Our central result is that retrieval gains vary systematically with the generator’s training regime rather than contributing a fixed additive improvement. Our 3D scaling law predicts held-out retrieval curves with a mean cross-validated relative error of 6.29% (Table 2). Extrapolation across model scales is slightly less reliable: mean LOMO error rises to 10.96%, driven primarily by extrapolation down to the 30M model (see Appendix A.5). On seven of nine benchmarks, the fits indicate larger retrieval gains for both smaller models and models with less pretraining per parameter. These patterns remain under other seeds, and we run a stability analysis in Appendix A.9.

To capture this dependence, we extend the 2D parametric law with interactions between retrieval gain, model capacity, and pretraining exposure:

$$L(N, D, R) = P(N, D) - C \left( \frac{N}{10^9} \right)^{-\delta} \cdots \left( \frac{D/N}{10} \right)^{-\zeta} \log \left( 1 + \eta \frac{R}{10^9} \right) \quad (2)$$

where  $N$  is model size (parameters),  $D$  is pretraining tokens, and  $R$  is retrieval/index tokens.  $P(N, D)$  is the parametric scaling baseline defined in Eq. 1. While we tested several formulations for incorporating the effects of retrieval, an interaction-based law most improved cross-validation error over an additive retrieval term; we report comparisons with nested and alternative functional forms in Appendix A.6. We evaluate predictive fit using five-fold grouped cross-validation, holding out entire  $(N, D)$  retrieval curves rather than individual datastore observations. Within each fold, the parametric baseline  $P(N, D)$  is refit using only the training-fold ( $R = 0$ ) configurations, ensuring that no information from a held-out model-pretraining configuration enters the fit.

Across the measured grid, a median 91% of the largest observed retrieval gain is present at  $R = N$ , followed by only small improvements as the datastore grows to as much as  $20N$ . Therefore, we observe early near-saturation of the benefit of retrieval over the  $R$  range we evaluate.

**The interaction depends on the evaluation target.** On five of six multiple-choice bench-

marks, choice-normalized Brier and decision-error fits agree in sign for both interaction exponents, whereas gold-answer likelihood and error agree on only one; their curve-bootstrap intervals are disjoint on all six (Appendix A.4). The reversal appears when competing answers are introduced: retrieval improves gold-answer likelihood more for smaller, less-pretrained models, but relative choice probability and accuracy more for larger, more-pretrained models. We therefore report them separately and restrict claims about downstream performance to the decision-level results.

## 4.3 Data Reuse

Most of retrieval’s measured benefit does not require novel information: it can be achieved from making information encountered during pretraining accessible again at inference. To separate these effects, we compare, for our larger models, retrieval from held-out documents with retrieval from documents in the model’s pretraining corpus. We decompose the held-out retrieval gain into an *access benefit*, measured by the improvement under reuse, and a *novelty premium*, measured by the additional improvement from held-out rather than reused data.

As shown in Figure 3, reuse accounts for 72–92% of the median held-out retrieval gain for models from 728M to 3B. Thus, train–index overlap reduces the magnitude of retrieval’s benefit but does not eliminate it. The novelty premium is largest in lightly pretrained regimes and generally approaches zero as  $D/N$  increases. Under our protocol, retrieval therefore acts primarily as an inference-time access mechanism, while access to unseen information provides an additional benefit primarily when pretraining is limited.

## 4.4 Pretraining–Retrieval Trade-off Curves

We now investigate the trade-off between pretraining data  $D$  and retrieval  $R$ , with the goal of understanding how retrieval can substitute for pretraining in reducing loss. Figure 4 summarizes this trade-off for two complementary perspectives: (i) the *substitutability* of retrieval for pretraining, and (ii) the *marginal benefit* of retrieval.

**Substitutability of retrieval.** For each model and pretraining scale, we fit scaling laws and compute the amount of retrieval required to match the performance of a baseline model trained without retrieval. We express retrieval in units of equivalent pretraining tokens as follows. For a configuration

|                    | Benchmark         | CV ARE<br>(%) | LOMO<br>(%) | $\alpha$ | $\beta$ | $C \times 10^3$ | $\delta$              | $\zeta$               | $L_0$ |
|--------------------|-------------------|---------------|-------------|----------|---------|-----------------|-----------------------|-----------------------|-------|
| Reason.            | CommonsenseQA     | 5.10          | 8.77        | 0.169    | 0.087   | 27.33           | -0.433 [-0.51, -0.33] | -0.247 [-0.30, -0.19] | 2.162 |
|                    | HellaSwag         | 1.76          | 3.31        | 0.476    | 0.505   | 0.13            | 1.295 [1.17, 1.44]    | 0.277 [0.26, 0.29]    | 1.484 |
|                    | StrategyQA        | 14.33         | 18.73       | 1.178    | 0.573   | 72.10           | -0.109 [-0.13, -0.09] | -0.236 [-0.25, -0.22] | 7.599 |
| Scientific/Open QA | SciQ              | 7.91          | 14.36       | 0.545    | 0.302   | 11.50           | 0.401 [0.20, 0.67]    | 0.072 [-0.05, 0.22]   | 0.952 |
|                    | OpenBookQA        | 3.14          | 7.13        | 0.376    | 0.239   | 4.60            | 0.731 [0.55, 0.97]    | 0.164 [0.04, 0.29]    | 1.952 |
|                    | AI2-ARC Easy      | 4.87          | 8.79        | 0.387    | 0.246   | 12.59           | 0.340 [0.28, 0.40]    | 0.101 [0.07, 0.13]    | 1.030 |
|                    | AI2-ARC Challenge | 3.68          | 5.84        | 0.303    | 0.293   | 8.24            | 0.469 [0.36, 0.57]    | 0.162 [0.09, 0.23]    | 1.113 |
|                    | Natural Questions | 5.61          | 12.76       | 0.391    | 0.392   | 13.21           | 0.604 [0.57, 0.63]    | 0.258 [0.23, 0.28]    | 1.259 |
|                    | SimpleQA          | 10.24         | 18.94       | 0.286    | 0.526   | 10.07           | 0.814 [0.79, 0.84]    | 0.388 [0.36, 0.41]    | 1.720 |
|                    | Mean              | 6.29          | 10.96       |          |         |                 |                       |                       |       |

Table 2: Per-benchmark fits of Eq. 2. **CV ARE** is the average relative error on held-out *whole*  $(N, D)$  retrieval curves, evaluated on retrieval points  $(R > 0)$ . **LOMO** is leave-one-model-size-out error.  $\delta$  and  $\zeta$  carry 95% bootstrap intervals obtained by resampling benchmark-evaluation noise;  $\delta > 0$  indicates larger absolute retrieval gains for smaller models and  $\zeta > 0$  larger gains in less-saturated pretraining regimes. Setting  $\delta = \zeta = 0$  recovers an additive retrieval law in which the gain is independent of  $N$  and  $D$ . The retrieval-rate parameter  $\eta$  reaches its upper bound of  $10^6$  on every benchmark, and profile fits continue improving or remain flat as this bound is increased. Thus, our datastore grid does not identify a finite retrieval-onset scale: the observed positive- $R$  response is already approximately linear in  $\log R$ .

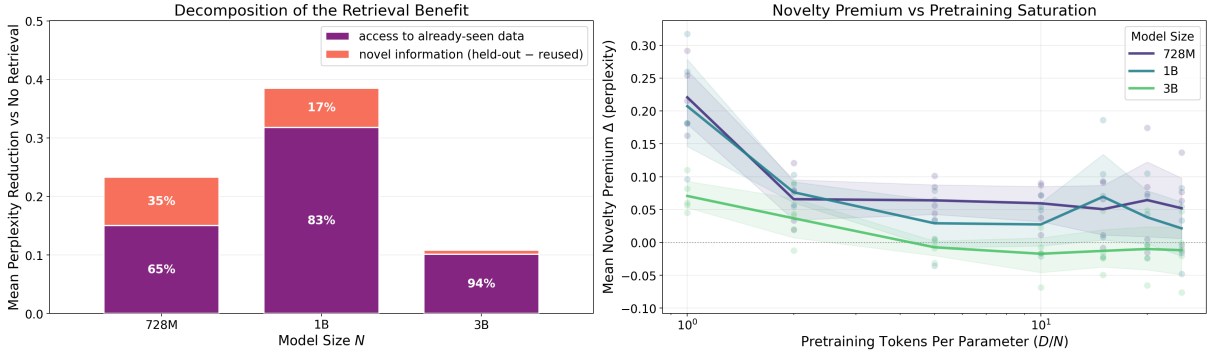


Figure 3: **Decomposing retrieval gains into reused and novel information.** *Left:* Mean perplexity reduction relative to no retrieval, decomposed into the benefit from re-accessing pretraining data and the additional novelty premium from using held-out data (HELD-OUT - REUSED). Labels report each component’s share of the total held-out retrieval gain. *Right:* The novelty premium as a function of pretraining tokens per parameter  $(D/N)$ . Points show benchmark-level values, lines show mean for each model size, and shaded regions show the bootstrapped 95% confidence interval.

$(N, D, R_{opt})$  with measured loss  $\mathcal{L}_{RAG}^*$ , we project this loss onto the  $N, D$  scaling curve (Eq. 1) to find the equivalent pretraining budget, giving us the projection  $D_{eff}^{RAG}$ . From this, we compute the *substitutability*  $\sigma$  which represents the number of pretraining tokens saved per retrieval token:

$$D_{eff}^{RAG} = \left( \frac{\mathcal{L}_{RAG}^* - \mathcal{L}_0 - A \cdot N^{-\alpha}}{B} \right)^{-1/\beta}$$

$$\Rightarrow \sigma = \frac{D_{eff}^{RAG} - D}{R_{opt}} \quad (3)$$

We observe a clear crossover behavior in Figure 4 (left). In low-data regimes, retrieval cannot effectively replace pretraining. However, it becomes increasingly effective beyond a median threshold

of  $\sim D/N = 5$  Pretraining Tokens Per Parameter across all model scales (estimated using the line-of-best-fit, with a 95% confidence interval of  $[2.9, 6.4]$ ), with each retrieval token replacing multiple pretraining ones. In this regime, the gains grow  $\sim \log$ -linearly, indicating that retrieval serves as an efficient alternative to additional pretraining. Importantly, this reflects *relative efficiency* rather than absolute improvement: even when retrieval substitutes efficiently for pretraining, the total achievable gain may be small if the baseline model is already near saturation.

**Marginal benefit of retrieval** is defined as the reduction in loss per unit of retrieval data  $\kappa = \Delta\mathcal{L}/(R/10^9)$ , where  $\Delta\mathcal{L} = \mathcal{L}_{R=0} - \mathcal{L}_{RAG}^*$  (higher

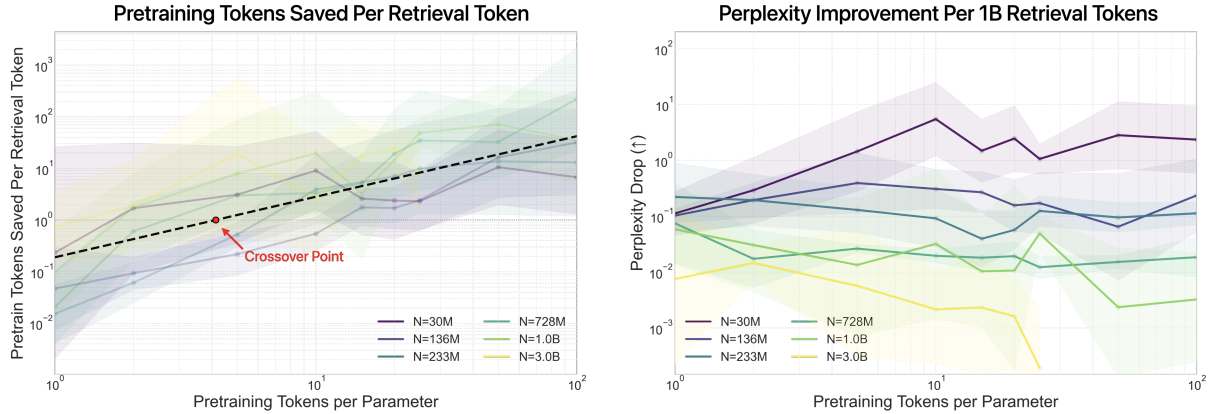


Figure 4: **Trade-off between pretraining and retrieval under a fixed data budget.** *Left:* We quantify the substitutability between retrieval and pretraining via the number of pretraining tokens saved per retrieval token, computed by fitting scaling laws and determining, for each pretraining scale, the amount of retrieval required to match baseline performance without retrieval. The dotted line represents a linear line-of-best-fit across all model scales. *Right:* We measure the marginal benefit of retrieval as perplexity improvement per billion retrieval tokens (higher is better) for models trained near their optimal pretraining ratio.

is better). Figure 4 (right) shows this quantity for models trained near their optimal pretraining ratio. We find that smaller models (e.g., 30M) benefit most from retrieval, achieving large improvements per unit of retrieved data. As model size increases, the marginal benefit decreases, with gains diminishing substantially at larger scales and largely saturating by 3B parameters. This suggests that while retrieval may remain an efficient substitute for pretraining at larger scales (left), the absolute improvement it provides diminishes as models become increasingly saturated. We note here that efficiency refers to performance per allocated corpus token, not end-to-end compute or deployment cost, for which Appendix A.3 reports a full accounting including storage and per-query inference overhead.

**Summary.** There clearly exists a scale-dependent trade-off between pretraining and retrieval. As model size and pretraining increase, retrieval’s marginal utility decreases, indicating a transition from retrieval-dominated to pretraining-dominated regimes. Most of the observed retrieval gain is already realized by  $R = N$ , and 72-92% remains when the datastore reuses pretraining data.

#### 4.5 Other Analyses

Because our fixed retrieval pipeline may limit absolute gains, Appendix A.10 examines alternative query formulations. Oracle-style, answer-aligned queries improve retrieval on knowledge-heavy tasks but produce little change on others, suggesting that retrieval quality shifts absolute performance without removing task-dependence.

## 5 Discussion

Our results show that retrieval and pretraining are coupled mechanisms for allocating access to corpus information. Our interaction scaling law captures this dependence: retrieval gains vary with model capacity and pretraining exposure rather than providing a fixed additive improvement. There is no metric-independent answer to which training regimes benefit most from retrieval. Retrieval improves the absolute likelihood of the gold continuation most for smaller, less-pretrained models, but improves its standing relative to competing choices, and ultimately decision accuracy, most for larger, more-pretrained models on the multiple-choice benchmarks studied. Thus, likelihood scaling characterizes how retrieved context changes support for the correct continuation, while decision-level scaling is the more relevant quantity if the objective is downstream accuracy. Gains are also strongly front-loaded: a median 91% of the largest observed improvement occurs by  $R = N$ , with limited returns from further datastore growth. Retrieval is therefore a task- and regime-dependent complement to parametric learning, not a uniform substitute for pretraining.

Our findings also clarify the role of retrieval quality. Improvements from better query formulation and oracle-style retrieval indicate that some of the observed trade-off is bottlenecked by the retriever rather than the language model. However, even stronger retrieval does not eliminate the need for parametric capacity, especially on reasoning-heavy tasks where the limiting factor appears to be com-



putation over knowledge. More broadly, this work suggests a shift in how pretraining corpora should be conceptualized. Rather than assuming that all available data should be compressed into weights, future language model design may benefit from explicitly partitioning corpora into data intended for internalization versus external access. This perspective aligns naturally with practical system design, where model capacity, training cost, memory footprint, and inference latency are all coupled.

**Limitations.** There are several factors in our present study that could be expanded. First, the retrieval setup is intentionally simple and fixed: we use a single retriever, a fixed chunking strategy, and a fixed top- $k$  protocol. Although this isolates the effect of retrieval scale, it likely understates the gains achievable with stronger retrieval pipelines, as we briefly investigate. Second, our evaluation focuses primarily on gold-answer perplexity as the most stable metric for scaling analysis; while appropriate for fitting smooth laws, this does not fully capture all downstream behaviors of interest. Third, although we study a broad range of model sizes, our conclusions are still limited to the scales, architectures, and corpora explored here.

**Future Work.** Future work can test whether reranking, learned filtering, adaptive chunking, or relevance-aware retrieval changes the interactions we identify. Additionally, we fit benchmark-specific scaling laws, but another direction is to identify latent structure that explains why some tasks benefit more from retrieval than others. This maybe via characterizing benchmarks by their knowledge dependence, retrieval sensitivity, or reasoning burden, and using these properties to build more general scaling laws over model size, pretraining, and external memory. Finally, inspired by human cognition, future work could explore purposeful allocation of abstract reasoning to pretraining vs. factual knowledge to retrieval.

## 6 Ethical Considerations

Our experiments require non-trivial computation for model pretraining, datastore embedding, and evaluation, with associated energy use and environmental impact. We limit experiments to models no larger than 3B parameters, reuse a fixed embedding model and precomputed embeddings across conditions, and construct nested datastores to avoid redundant processing. Appendix A.3 provides an

estimated accounting of these computational costs. Additionally, our models and retrieval stores are derived from DCLM web data, which may contain social biases, offensive material, personal information, or copyrighted text. Retrieval can make such source content more directly accessible at inference time. Our experiments study aggregate scaling behavior rather than deploying a user-facing system, but these risks should be considered before applying the resulting methods in practice.

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## A Appendix

### A.1 Pretraining setup

All pretraining runs use NVIDIA H100 GPUs with 8 devices per job. The total expenditure across training, retrieval embedding and index creation, and evaluation was approximately 15,000 GPU hours. We train with FSDP data-parallelism, in mixed precision, and without model parallelism. We use varied per-device micro-batch sizes depending on the model scale, and gradient accumulation to achieve an effective global batch size of 256 across runs. Training uses context length (block size) 4096 across all models. Additional hyperparameter details for each model scale is provided in Table 6.

### A.2 Index Construction

#### A.2.1 Retrieval Corpus

We construct a retrieval corpus from a held-out split of the DCLM dataset, chunked into overlapping token windows and embedded using a pretrained embedding model. All embeddings are  $L2$ -normalized and indexed using FAISS with product quantization. We vary the retrieval corpus size across multiple scales and construct separate indices for each setting. Each split corresponds to an increasing retrieval corpus size derived from disjoint subsets of the DCLM corpus. Index construction-relevant details are in Tables 3, 4, and 5.

#### A.2.2 Embedding Choice, Chunking & Tokenization

We explored several embedding models including BAAI/bge-base-en-v1.5, google/embeddinggemma-300m, and Qwen3-Embedding-8B. We chose Qwen3-Embedding-8B as it showed strongest semantic recall and was consistently at the top of public RAG benchmarks. We used IVFPQ for our indexing algorithm, a chunk length of 900 tokens, and a stride length of 256 tokens. We built our chunks using TikToken cl100k-base (OpenAI, 2022), then decoded back to text. Tables 3, 4, and 5 describe our specific embedding, chunking, and FAISS configurations.

### A.3 Cost Accounting

We estimate generator pretraining as  $F_{\text{train}} \approx 6ND$  and one-time datastore embedding as  $F_{\text{embed}} \approx 2N_eR$ , where  $N_e = 8B$  for Qwen3-Embedding-8B. Table 7 reports these costs at the largest pretraining and retrieval scales evaluated for

| Component           | Configuration                     |
|---------------------|-----------------------------------|
| Embedding model     | Qwen3-Embedding-8B                |
| Embedding dimension | 4096                              |
| Pooling             | Last non-padded token             |
| Normalization       | $L2$ -normalized                  |
| Tokenizer           | cl100k_base                       |
| Chunk length        | 900 tokens                        |
| Stride              | 256 tokens ( $\sim 28\%$ overlap) |

Table 3: Embedding and chunking configuration for retrieval corpus construction.

| Component         | Configuration                        |
|-------------------|--------------------------------------|
| Index type        | IVFPQ (inner product)                |
| Sub-quantizers    | 128                                  |
| Bits per code     | 8                                    |
| Search parameter  | nprobe = 64                          |
| Training set size | $\max(\text{nlist} \times 50, 10^5)$ |
| Batch size (add)  | 500K vectors                         |

Table 4: FAISS index configuration.

| Corpus Split | nlist  | Shards |
|--------------|--------|--------|
| 5%           | 32,768 | 32     |
| 10%          | 32,768 | 64     |
| 20%          | 65,536 | 128    |
| 30%          | 65,536 | 128    |

Table 5: Per-split FAISS index configuration.

each generator. Because the datastores are nested, embedding cost is charged once for the largest store rather than separately for every datastore size.

Embedding is therefore more expensive than generator pretraining for the 30M–233M models, but falls to  $0.73\times$ ,  $0.53\times$ , and  $0.24\times$  of pretraining cost for the 728M, 1B, and 3B models. The fixed embedder does not confound comparisons among datastore sizes for a given generator, but its absolute cost is non-negligible, particularly for small generators.

Retrieval also introduces storage and per-query overhead. Under our IVFPQ configuration, the compressed vector codes for a 20B-token store require approximately 3.0GB, plus approximately 1.1GB for coarse centroids; storing the associated tokenized text can require up to 80GB. For a representative 128-token query, query embedding costs approximately  $2.05 \times 10^{12}$  FLOPs. Retrieving five 900-token chunks can fill the generator’s 4,096-token context window, adding approximately  $1.02 \times 10^{13}$  and  $2.83 \times 10^{13}$  prefill FLOPs for the 1B and 3B generators, respectively. At one million queries, query embedding and additional prefill amount to approximately 2.0% and 2.2% of their

| Model Size | Layers | Hidden Dim | Number of Heads | Query Groups | Intermediate Size | Block Size |
|------------|--------|------------|-----------------|--------------|-------------------|------------|
| 30M        | 8      | 256        | 4               | 4            | 512               | 4096       |
| 136M       | 8      | 512        | 8               | 8            | 2048              | 4096       |
| 233M       | 16     | 640        | 10              | 10           | 2560              | 4096       |
| 728M       | 18     | 1280       | 10              | 10           | 5120              | 4096       |
| 1B         | 24     | 1408       | 11              | 11           | 5632              | 4096       |
| 3B         | 26     | 2560       | 20              | 20           | 10240             | 4096       |

Table 6: Pretraining hyperparameters across model sizes

| Generator | $D_{\max}$<br>(B tokens) | $R_{\max}$<br>(B tokens) | $F_{\text{train}}$<br>(FLOPs) | $F_{\text{embed}}$<br>(FLOPs) | Embed / Train |
|-----------|--------------------------|--------------------------|-------------------------------|-------------------------------|---------------|
| 30M       | 4.50                     | 0.60                     | $8.10 \times 10^{17}$         | $9.60 \times 10^{18}$         | 11.85         |
| 136M      | 20.40                    | 2.72                     | $1.66 \times 10^{19}$         | $4.35 \times 10^{19}$         | 2.61          |
| 233M      | 34.95                    | 4.66                     | $4.89 \times 10^{19}$         | $7.46 \times 10^{19}$         | 1.53          |
| 728M      | 72.80                    | 14.56                    | $3.18 \times 10^{20}$         | $2.33 \times 10^{20}$         | 0.73          |
| 1B        | 100.00                   | 20.00                    | $6.00 \times 10^{20}$         | $3.20 \times 10^{20}$         | 0.53          |
| 3B        | 75.00                    | 20.00                    | $1.35 \times 10^{21}$         | $3.20 \times 10^{20}$         | 0.24          |

Table 7: Approximate generator-pretraining and one-time datastore-embedding costs at the largest evaluated scales. Estimates use  $6ND$  for training and  $2N_e R$  for embedding with  $N_e = 8B$ .

largest pretraining runs.

#### A.4 Choice-Normalized and Accuracy Fits

Our primary scaling laws use gold-answer perplexity, which measures the model’s absolute support for the correct continuation without reference to competing choices. To determine whether the resulting interactions transfer to downstream decisions, we refit Eq. 2 using two additional targets on the six multiple-choice benchmarks: choice-normalized Brier loss and decision error. Brier loss is computed after normalizing likelihoods across the answer choices, while decision error applies the final argmax. These targets form a ladder from absolute gold-answer support, through competition among choices, to the discrete prediction.

The decision-error law achieves (6.30%) mean CV ARE, comparable to the perplexity law’s (6.29%) (Table 8). However, both interaction exponents are negative on all six accuracy benchmarks, indicating larger decision-level gains for larger and more heavily pretrained models. This reverses the likelihood pattern on five of the six shared benchmarks. We omit benchmarks without stable decision-level fits. Natural Questions and SimpleQA have nearly constant exact-match error, while StrategyQA drives interaction parameters to their bounds.

Figure 5 localizes where this reversal appears. For both exponents, Brier and decision-error fits agree in sign on five of six benchmarks, whereas

perplexity and decision error agree on only one. The perplexity and decision-error intervals are disjoint on all six benchmarks for both exponents, indicating that the difference is not explained by uncertainty under the curve-resampling procedure. The interaction therefore changes when competing answers enter the evaluation, before the final argmax: retrieval improves absolute gold-answer likelihood most for smaller, less-pretrained models, but improves the gold answer relative to its competitors most for larger, more-pretrained models. The metrics therefore characterize distinct scaling targets. Gold-answer perplexity measures how retrieved context changes support for the correct continuation; decision errors measure whether that support improves relative to alternative answers.

#### A.5 Leave-One-Model-Out Errors

Leave-one-model-out (LOMO) validation removes one model size entirely, refits the law (both the parametric term and the retrieval term) on the remaining five, and predicts the held-out size’s retrieval curves. It therefore measures extrapolation across scales, unlike the grouped cross-validation of Table 2, which holds out whole  $(N, D)$  retrieval curves but never an entire model size.

As Table 9 shows, LOMO error is concentrated in the 30M fold. Withholding 30M produces a 33.4% mean error, compared with 5.3-8.0% for every other held-out size. Excluding this fold reduces the mean to 6.1%, close to the grouped-CV error.

|               | Benchmark         | CV ARE (%) | LOMO (%) | $\alpha$ | $\beta$ | $C \times 10^3$ | $\delta$                        | $\zeta$                 | $L_0$ |
|---------------|-------------------|------------|----------|----------|---------|-----------------|---------------------------------|-------------------------|-------|
| Reas.         | CommonsenseQA     | 4.10       | 10.53    | 0.157    | 0.098   | 1.06            | $-1.462 [-1.65, -1.30]$         | $-0.699 [-0.80, -0.60]$ | 0.255 |
|               | HellaSwag         | 4.16       | 7.81     | 0.100    | 0.069   | 0.40            | $-2.000^\dagger [-2.00, -2.00]$ | $-0.924 [-0.97, -0.88]$ | 0.262 |
| Scientific QA | SciQ              | 17.30      | 51.96    | 0.618    | 0.359   | 0.34            | $-1.216 [-1.51, -0.99]$         | $-0.751 [-0.97, -0.62]$ | 0.046 |
|               | OpenBookQA        | 2.23       | 4.95     | 0.068    | 0.047   | 1.69            | $-1.112 [-1.49, -0.85]$         | $-0.595 [-0.80, -0.42]$ | 0.364 |
|               | AI2-ARC Easy      | 7.27       | 20.32    | 0.259    | 0.171   | 2.32            | $-0.894 [-0.98, -0.82]$         | $-0.474 [-0.52, -0.43]$ | 0.148 |
|               | AI2-ARC Challenge | 2.74       | 5.85     | 0.056    | 0.036   | 0.99            | $-1.524 [-1.75, -1.34]$         | $-0.817 [-0.94, -0.71]$ | 0.328 |
| Mean          |                   | 6.30       | 16.90    |          |         |                 |                                 |                         |       |

Table 8: Eq. 2 refit to **downstream error rate** ( $L = 1 - \text{accuracy}$ ) instead of gold-answer perplexity, on the six multiple-choice benchmarks for which accuracy is defined and produces stable fits.

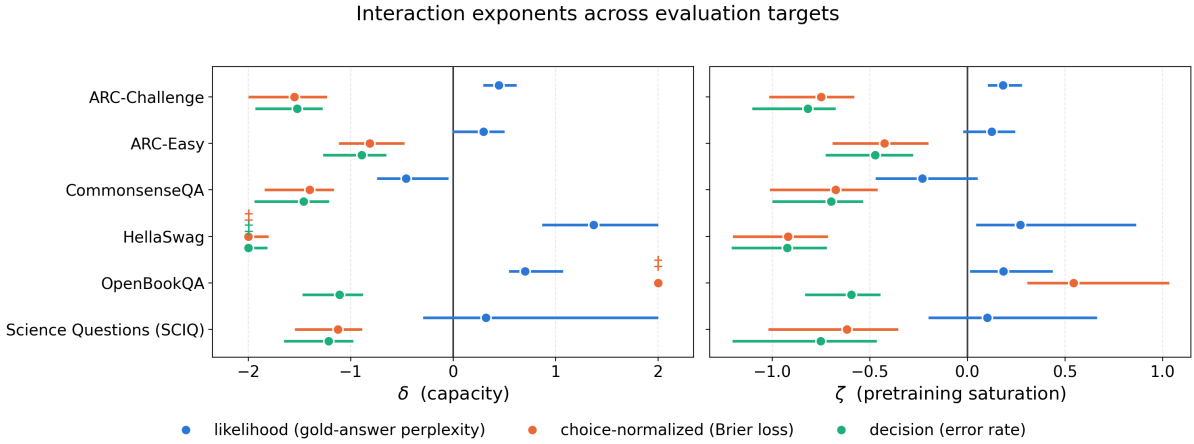


Figure 5: **Interaction exponents across evaluation targets.** Fitted capacity exponent  $\delta$  (left) and pretraining-exposure exponent  $\zeta$  (right) under gold-answer perplexity, choice-normalized Brier loss, and decision error. Positive values indicate larger retrieval gains for smaller models ( $\delta$ ) or models with fewer pretraining tokens per parameter ( $\zeta$ ). Bars show 95% intervals obtained by resampling complete  $(N, D)$  retrieval curves. Estimates at  $\pm 2$  lie on an optimization bound.

This fold uniquely requires downward extrapolation from models spanning 136M–3B. Under the power-law baseline, uncertainty in the capacity exponent is amplified toward smaller ( $N$ ), while the fit must also predict the large retrieval responses observed at 30M. The resulting errors are concentrated on StrategyQA (93.5%), Natural Questions (54.1%), and SimpleQA (46.3%), which exhibit the noisiest behavior at the smallest scale.

By contrast, withholding the upper boundary at 3B produces the lowest mean error, 5.3%. The elevated aggregate LOMO error therefore reflects difficulty extrapolating below the observed range rather than uniformly weak transfer across model sizes. Within the scales studied, interpolation and upward prediction to 3B are comparable in accuracy to predicting held-out retrieval curves.

## A.6 Additional Forms

We compare alternative retrieval laws to determine whether retrieval is additive, which aspects

of the training regime modulate its benefit, and whether the observed datastore response supports a more flexible functional form. All models share the same parametric baseline  $P(N, D)$ , satisfy  $G(R = 0) = 0$ , and are evaluated using the grouped cross-validation protocol described in Section 4.2.

Table 10 first establishes that a retrieval term is necessary: the no-retrieval model has substantially higher predictive error than the additive law  $M_0$ . Allowing retrieval magnitude to depend on model capacity alone reduces mean CV ARE from 7.04% to 6.67%, whereas conditioning only on pretraining saturation yields a smaller reduction to 6.98%. Conditioning on both produces the strongest nested model,  $M_{NS}$ , which reaches 6.29% CV ARE and improves over  $M_0$  on all benchmarks. This comparison supports model capacity as the dominant interaction, with  $D/N$  providing a weaker complementary effect.

More flexible laws do not yield a clear improve-

| Benchmark         | 30M  | 136M | 233M | 728M | 1B   | 3B   | All folds | Excl. 30M |
|-------------------|------|------|------|------|------|------|-----------|-----------|
| CommonsenseQA     | 19.6 | 4.4  | 3.7  | 4.1  | 5.2  | 15.5 | 8.8       | 6.6       |
| HellaSwag         | 10.2 | 1.5  | 2.6  | 1.7  | 1.7  | 2.7  | 3.4       | 2.1       |
| StrategyQA        | 93.5 | 16.3 | 14.4 | 17.3 | 7.2  | 7.6  | 26.1      | 12.6      |
| SciQ              | 28.8 | 10.3 | 8.9  | 7.1  | 7.4  | 2.7  | 10.9      | 7.3       |
| OpenBookQA        | 18.4 | 2.7  | 3.9  | 3.8  | 4.1  | 5.6  | 6.4       | 4.0       |
| AI2-ARC Easy      | 18.4 | 8.1  | 4.7  | 4.1  | 6.7  | 3.1  | 7.5       | 5.4       |
| AI2-ARC Challenge | 11.7 | 4.7  | 2.7  | 3.1  | 4.5  | 3.0  | 4.9       | 3.6       |
| Natural Questions | 54.1 | 7.2  | 3.4  | 4.5  | 6.8  | 4.5  | 13.4      | 5.3       |
| SimpleQA          | 46.3 | 16.4 | 7.4  | 5.2  | 10.2 | 3.4  | 14.8      | 8.5       |
| Mean              | 33.4 | 8.0  | 5.8  | 5.7  | 6.0  | 5.3  | 10.7      | 6.1       |

Table 9: LOMO average relative error (%) on held-out retrieval points ( $R > 0$ ), by which model size was withheld.

ment in out-of-scale prediction. Allowing  $N$  and  $D/N$  to modulate the retrieval rate lowers interpolation error, as do Hill laws with a varying transition point, but both increase LOMO error relative to  $M_{NS}$ . Their rate or transition parameters also lie on optimization bounds in nearly every benchmark. The bounded Hill interaction  $H_{NS}$  performs nearly identically to the logarithmic  $M_{NS}$ , showing that the magnitude-interaction result is not specific to the logarithmic response. Replacing absolute datastore size with  $R/N$  also leaves both CV and LOMO error essentially unchanged. We select  $M_{NS}$  as the primary law because it captures the supported interaction with fewer assumptions than the rate-varying alternatives.

### A.7 Some Further Notes On Retrieval Efficiency Metrics ( $\sigma$ and $\kappa$ )

To complement the scaling-law analysis in §4.4, we introduce two derived metrics that quantify the efficiency of retrieval relative to pretraining: **replacement cost** ( $\sigma$ ) and **marginal benefit** ( $\kappa$ ). These metrics provide an interpretable view of the trade-off between parametric and non-parametric knowledge. We explain them further here.

**Multiplicative nature of  $\sigma$ .** Replacement cost  $\sigma$  is a ratio that measures how many pretraining tokens are replaced per retrieval token. As such, it is inherently multiplicative: a change from  $\sigma = 1$  to  $\sigma = 10$  represents a tenfold increase in efficiency.

Consistent with this interpretation,  $\sigma$  spans multiple orders of magnitude across tasks (e.g., from  $< 1$  to  $> 10^3$ ), and follows approximately log-linear trends with respect to pretraining scale. Therefore, we aggregate  $\sigma$  using the geometric mean:

$$\sigma_{\text{GM}} = \exp\left(\frac{1}{n} \sum_{i=1}^n \ln \sigma_i\right) \quad (4)$$

This preserves multiplicative structure, reduces sensitivity to extreme values, and aligns with the power-law scaling behavior underlying our analysis.

**Additive nature of  $\kappa$ .** In contrast, the marginal benefit  $\kappa$  measures an absolute reduction in loss per unit of retrieval data. This is an additive quantity: improvements combine linearly and may be positive, zero, or negative depending on the task.

Because  $\kappa$  can take non-positive values and does not exhibit multiplicative structure, geometric aggregation is not appropriate. Instead, we summarize  $\kappa$  using the median:

$$\kappa_{\text{med}} = \text{median}(\{\kappa_1, \dots, \kappa_n\}) \quad (5)$$

which provides a robust estimate of the typical improvement across tasks.

**Pretraining Regimes.** We report results across three regimes defined by the token-to-parameter ratio:

- $1\times$ :  $D/N \approx 1$  (undertrained)
- $10\times$ :  $D/N \approx 10$  (near-optimal)
- $100\times$ :  $D/N \approx 100$  (overtrained)

#### A.7.1 Benchmark-Level Results

Table 11 shows  $\sigma$  and  $\kappa$  at various training scales for each benchmark.

#### Observations:

- **Retrieval substitutability increases with saturation:**  $\sigma$  grows with  $D$ , indicating that retrieval becomes more effective once pretraining enters diminishing returns.
- **Diminishing marginal returns:**  $\kappa$  decreases with model size, consistent with larger models internalizing more knowledge parametrically.



| Model         | Retrieval gain $G$                                          | #par. | CV ARE (%)  | $\Delta$ (%) | wins       | LOMO (%)     | $\Delta$ (%) |
|---------------|-------------------------------------------------------------|-------|-------------|--------------|------------|--------------|--------------|
| $M_0$         | 0 (no retrieval term)                                       | 0     | 9.24        | +32.5        | 0/9        | 14.91        | +18.1        |
| $M_0$         | $C \log(1 + \eta r)$                                        | 2     | 7.04        | —            | —          | 12.85        | —            |
| $M_N$         | $C n^{-\delta} \log(1 + \eta r)$                            | 3     | 6.67        | -7.2         | 8/9        | 11.36        | -12.7        |
| $M_S$         | $C s^{-\zeta} \log(1 + \eta r)$                             | 3     | 6.98        | -1.4         | 7/9        | 12.82        | -0.5         |
| $M_{NS}$      | $C n^{-\delta} s^{-\zeta} \log(1 + \eta r)$                 | 4     | <b>6.29</b> | <b>-12.7</b> | <b>9/9</b> | <b>10.96</b> | <b>-16.3</b> |
| $M_{NS,\eta}$ | $C n^{-\delta} s^{-\zeta} \log(1 + \eta n^{-u} s^{-v} r)$   | 6     | 6.06        | -16.3        | 9/9        | 11.42        | -12.0        |
| $H_{NS}$      | $C n^{-\delta} s^{-\zeta} r / (r + K)$                      | 4     | 6.30        | -12.7        | 9/9        | 11.23        | -13.8        |
| $H_{NS,K}$    | $C n^{-\delta} s^{-\zeta} r / (r + K_0 n^\kappa s^\lambda)$ | 6     | 5.94        | -17.8        | 9/9        | 11.50        | -10.0        |
| $M_{NS}(R/N)$ | as $M_{NS}$ , with $r \rightarrow R/N$                      | 4     | 6.28        | -12.9        | 9/9        | 10.94        | -16.1        |

Table 10: **Comparison of retrieval-gain specifications.** Results are averaged across the benchmarks. All specifications share the same five-parameter baseline  $P(N, D)$ .  $\Delta$  is the relative change from the additive law  $M_0$  and **wins** counts benchmarks with lower grouped-CV error than  $M_0$ .

| Benchmark          | Mean $\sigma$ | $\sigma@1\times$ | $\sigma@10\times$ | $\sigma@100\times$ | Mean $\kappa$ | $\kappa@1\times$ | $\kappa@10\times$ | $\kappa@100\times$ |
|--------------------|---------------|------------------|-------------------|--------------------|---------------|------------------|-------------------|--------------------|
| AI2C-ARC Challenge | 4.73          | 0.06             | 9.14              | 11.20              | 0.4778        | 0.1582           | 2.4505            | 0.4216             |
| AI2C-ARC Easy      | 22.54         | 0.38             | 27.83             | 64.40              | 0.3701        | 0.1048           | 0.4348            | 0.5736             |
| CommonsenseQA      | 677.51        | 677.93           | 1216.95           | 1951.00            | 0.5738        | 0.2715           | 1.4521            | 1.0438             |
| NQ-Open            | 5.28          | 0.23             | 2.72              | 13.63              | 0.4421        | 0.4118           | 0.5427            | 0.3192             |
| OpenBookQA         | 6.65          | 0.32             | 6.05              | -10.49             | 2.2450        | 0.2014           | 3.8579            | 0.4100             |
| SciQ               | 24.21         | 0.20             | 11.69             | 78.62              | 0.9830        | 0.3722           | 0.2576            | 0.4918             |
| StrategyQA         | 95.59         | 7.89             | 0.95              | 105.40             | 3.5006        | 0.0420           | 0.3923            | 2.1839             |

Table 11: Replacement cost  $\sigma$  and marginal benefit  $\kappa$  across benchmarks and training regimes.

- **Strong task dependence:** knowledge-intensive tasks (e.g., CommonsenseQA) exhibit very high  $\sigma$ , while reasoning-heavy tasks (e.g., HellaSwag) show weak or negative gains.

These results provide a quantitative interpretation of the trade-off curves in Figure 4, reinforcing the view that retrieval acts as a scale-dependent substitute for pretraining.

## A.8 Calibration Plots

Calibration plots (Figure 10) allow us to visualize how our predicted loss (from the 3D scaling law fit) compares to the actual loss values.

## A.9 Stability Analysis

To assess the robustness of our scaling-law fits, we evaluate stability across multiple random seeds and model initializations. Specifically, we consider three random seeds each for three model families (30M, 136M, and 233M), yielding a total of 27 runs. For each configuration, we fit scaling laws independently and compute both cross-validation average relative error (CV ARE) and leave-one-model-out ARE (LOMO ARE). Figure 6 reports the mean and standard deviation of these metrics across runs for each benchmark. Overall, we observe low variance

in both CV ARE and LOMO ARE across most tasks, indicating that the fitted scaling relationships are stable with respect to initialization and data ordering. Reasoning-heavy tasks such as StrategyQA exhibit higher variance and larger absolute errors, suggesting that their scaling behavior is noisier and less well captured by simple parametric forms. In contrast, more knowledge-driven benchmarks (e.g., ARC, OpenBookQA) show consistently low variance and strong fit quality across runs.

## A.10 RAG Improvements

While our primary focus is on the allocation trade-off between pretraining and retrieval, this raises a complementary question: how much of the observed benefit depends on retrieval quality? Qualitatively, retrieved contexts on factoid QA often capture the correct topic but do not consistently contain directly answer-bearing evidence (e.g., specific entities or dates), suggesting that retrieval precision remains a limiting factor. To probe this further, we evaluate a simple strategy for improving retrieval: varying retriever query formulation. We compare, with a fixed corpus index across methods, (i) question-only queries, (ii) queries augmented with answer choices (when applicable), and (iii)

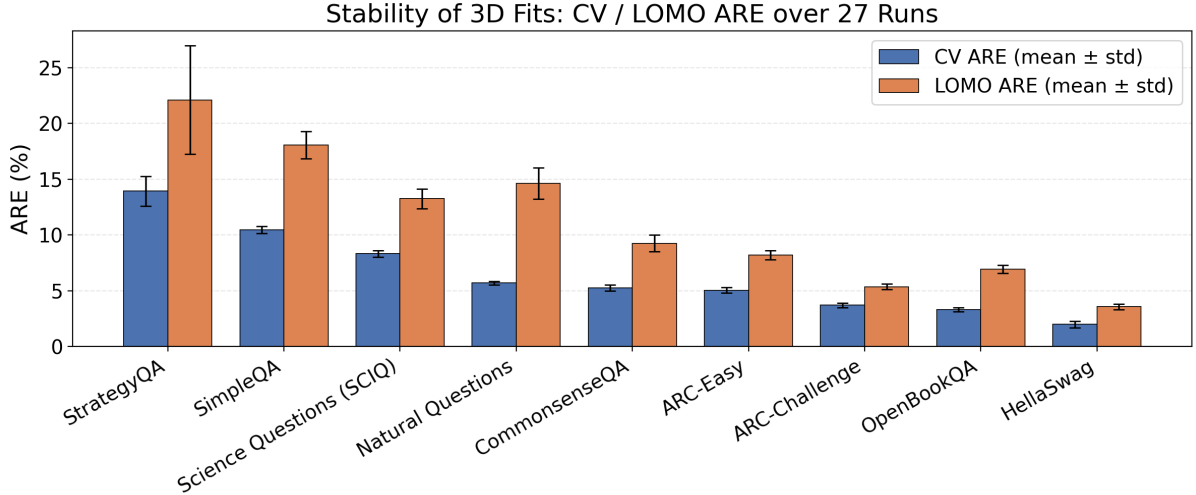


Figure 6: **Stability of scaling-law fits across random seeds.** We report mean  $\pm$  standard deviation of cross-validation ARE (CV ARE) and leave-one-model-out ARE (LOMO ARE) across 27 separate fits (every possible combination of 3 model families  $\times$  3 seeds each).

queries that include the gold answer (an oracle).<sup>1</sup>

All RAG conditions use the same FAISS index, constructed from a held-out corpus whose size is equivalent to 20% of the maximum pretraining-token budget. We prepend the top- $k = 5$  retrieved passages to the prompt and sweep pretraining tokens per parameter while keeping the retrieval configuration fixed.

Figure 7, 8, and 9 report results on SimpleQA, LAMBADA GSM8K, and CommonsenseQA. Retrieval yields modest gains on knowledge-heavy tasks (SimpleQA, CommonsenseQA), particularly when queries better align with the answer, with improvements increasing at larger model scales. In contrast, reasoning-heavy tasks (GSM8K (Cobbe et al., 2021), LAMBADA (Paperno et al., 2016)<sup>2</sup>), show minimal change. Across tasks, improved retrieval yields incremental gains but does not alter the scaling trends observed earlier. This reinforces prior takeaways that retrieval is not a uniform substitute for pretraining, and that its effectiveness depends on both model scale and task type.

<sup>1</sup>This is closer to an approximate upper bound by more closely approximating an optimal scenario where one would have a (near) perfect retriever to maximize the potential benefits of RAG.

<sup>2</sup>We try RAG improvements on these two additional benchmarks (math and word prediction) that performed poorly with retrieval on our initial pilot studies.

### A.11 Qualitative Analysis of Retrieval Behavior

To better understand when retrieval helps, we manually inspected retrieved contexts for a couple of datasets where RAG helps (SimpleQA, NQ-Open) and one where it does not (GSM8K). The key contrast is that retrieval is often useful as *topical grounding* for SimpleQA and NQ-Open, but is much less useful for GSM8K, where problems are typically self-contained and external text is often distractive.

For SimpleQA, query-only contexts often match the right domain (e.g., history, philosophy, astronomy, biology) and sometimes surface source-adjacent material that can support answer extraction. For example, legal-history questions about *A Survey of London* retrieve medieval court/tower records; philosophy questions about Hegel retrieve prose that discusses his views on Romantic art and its main art forms; and technical biomedical questions retrieve mutation/gene-focused research text rather than generic web chatter.

For NQ-Open, RAG helps mostly when retrieval injects a single concrete anchor (year/number/entity). For example, “*when was the first Australian prime minister elected*” changes from baseline 1977 to correct 1901 with RAG, and “*what age do you need to be to buy a BB gun*” shifts from 5 to 18 years old after retrieval includes age-threshold text (“*over the age of 18... over 21*”).

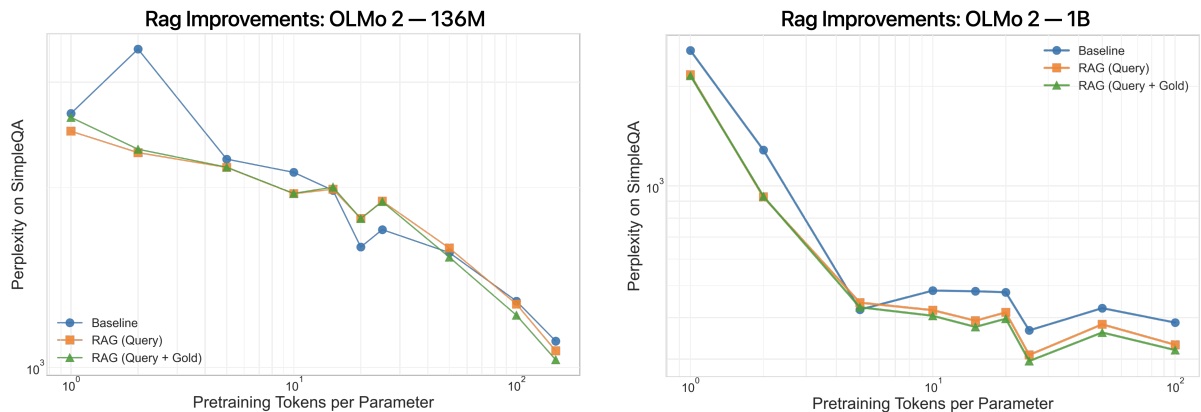


Figure 7: **Effect of retrieval query formulation on performance.** Comparison of standard generation on SimpleQA without retrieval (Baseline) to RAG under two query formulations: (i) *RAG (Query)*, which retrieves top- $k$  passages using only the question, and (ii) *RAG (Query + Gold)*, which includes the gold answer in the query too (an oracle-style ablation). SimpleQA is not multiple-choice (no answer choices), so we do not report *RAG (Query + Choices)* here. All methods use a shared corpus index constructed from 20% of the data, retrieving the top-5 passages per query. *Left*: OLMo-2 136M. *Right*: OLMo-2 1B.

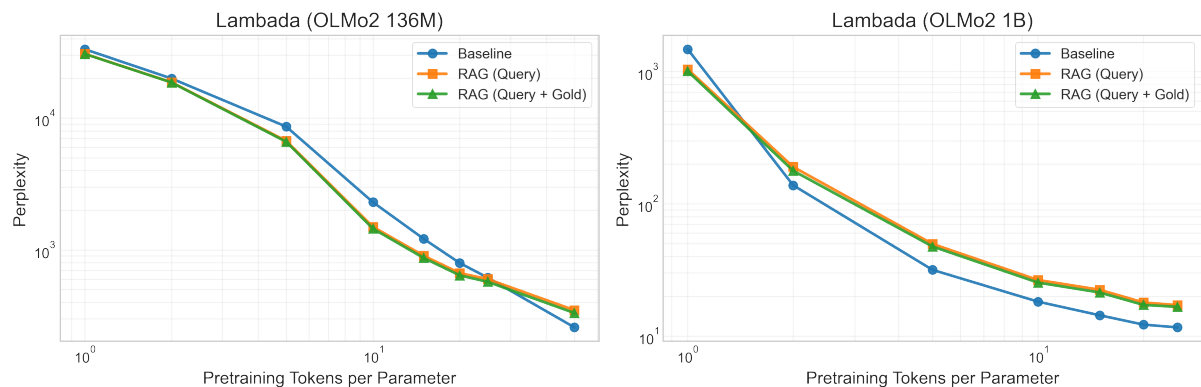


Figure 8: **Effect of retrieval query formulation on LAMBADA.** Similar to Figure 7 above.

for handguns”). Likewise, “when was the last time anyone was on the moon” moves from “200 years ago” to 1972 once Apollo timeline snippets appear. Sometimes the same mechanism still misses the target: retrieval changes the model’s answer, but the new answer remains incorrect. Overall, the benefit is best characterized as occasional fact anchoring from salient cues, rather than consistently reliable evidence grounding.

For GSM8K, query-only retrieval is usually not needed and often noisy. Retrieved passages are commonly worksheets, forum posts, product listings, dictionary pages, or malformed index fragments (e.g., “files\_...”), which rarely contribute to the arithmetic decomposition required by the question. Occasionally, retrieval provides a useful conversion fact (e.g., *gallon-to-pints*), but most examples are either redundant with what is already stated in the prompt or off-task. This qualitative pattern

aligns with the quantitative result that RAG yields little benefit on GSM8K in this setup, and with existing conclusions that RAG helps more heavily with long-tail factual knowledge than things like mathematics.

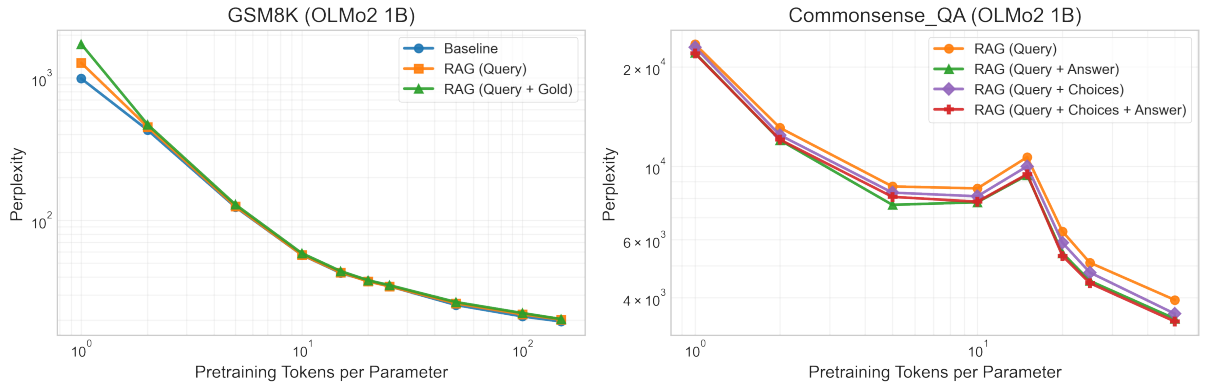


Figure 9: **Effect of retrieval query formulation across GSM8K and CommonsenseQA.** Similar to Figure 7. For CommonsenseQA, we additionally compare querying both choices and gold answer (RAG Query + Choices + Answer). Both panels show OLMo-2 1B as a function of pretraining tokens per parameter.



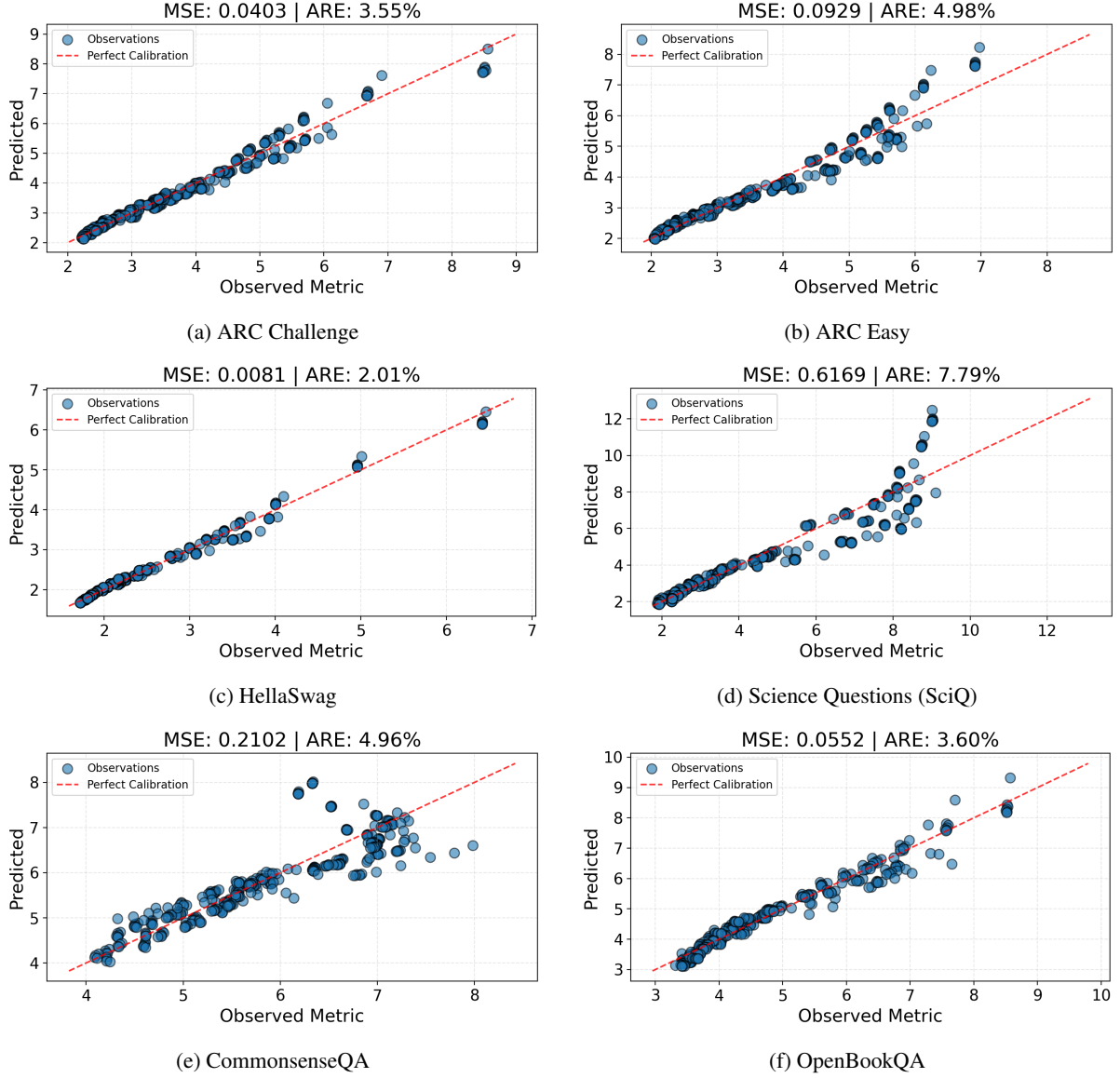


Figure 10: **Calibration plots for 3D scaling law fits across benchmarks.** We show the alignment between predicted and observed gold-answer perplexity across six benchmarks for the interaction model  $\mathcal{M}_{NS}$  (Eq. 2), which augments the Hoffmann power-law formulation with a capacity- and saturation-modulated logarithmic retrieval term. The tight grouping around the diagonal indicates that the model effectively captures retrieval-augmented scaling behavior. Reported MSE and ARE values are computed in-sample over the plotted points; held-out cross-validation errors are reported separately in Table 2.